

ARCHITECTURE DECISION RECORD (ADR)

CivicPulse

Version: 1.0

Status: Accepted

ADR-001

Use Modular Monolith Instead of Microservices

Status

Accepted

Context

Tim pengembang kecil.

Produk masih MVP.

Belum ada kebutuhan skalabilitas ekstrem.

Decision

Menggunakan Modular Monolith Architecture.

Consequences

Pros

- Development lebih cepat
- Deployment lebih sederhana
- Maintenance lebih mudah
- Cocok untuk MVP

Cons

- Perlu disiplin menjaga boundaries antar module
-

ADR-002

Use Next.js 16 Full Stack

Status

Accepted

Context

Frontend dan backend membutuhkan integrasi cepat.

Decision

Menggunakan Next.js App Router sebagai Full Stack Framework.

Consequences

Pros

- Single codebase
- SSR dan SEO
- Type sharing
- Faster development

Cons

- Vendor lock terhadap ecosystem React
-

ADR-003

Use PostgreSQL As Primary Database

Status

Accepted

Context

Mebutuhkan relasi kompleks.

Decision

PostgreSQL sebagai database utama.

Consequences

Pros

- Relational
- ACID compliant
- Analytics friendly
- Mature ecosystem

Cons

- Lebih kompleks dibanding SQLite
-

ADR-004

Use Prisma ORM

Status

Accepted

Decision

Prisma sebagai ORM.

Consequences

Pros

- Type-safe
- Excellent DX
- Migration support

Cons

- Query kompleks perlu optimization
-

ADR-005

Use Better Auth

Status

Accepted

Decision

Better Auth untuk authentication.

Consequences

Pros

- Session based
- Modern
- Next.js friendly

Cons

- Community lebih kecil dibanding NextAuth
-

ADR-006

Use RBAC Authorization

Status

Accepted

Roles

CITIZEN

OFFICER

SUPER_ADMIN

Reason

Mempermudah pengelolaan hak akses.

ADR-007

Use Service Layer Pattern

Status

Accepted

Decision

Business logic tidak boleh berada di route handler.

Flow

Route

↓

Service

↓

Repository

↓

Database

Reason

Maintainability.

Testability.

Scalability.

ADR-008

Use Repository Pattern

Status

Accepted

Decision

Semua query database harus melalui repository.

Reason

Memisahkan business logic dan data access.

ADR-009

Use Redis For Cache

Status

Accepted

Decision

Redis digunakan untuk:

- Dashboard cache
 - Analytics cache
 - Session optimization
-

ADR-010

Use BullMQ For Background Jobs

Status

Accepted

Decision

Semua asynchronous task diproses melalui queue.

Examples

- Email
 - Notifications
 - Analytics refresh
-

ADR-011

Use S3 Compatible Storage

Status

Accepted

Decision

Attachment tidak disimpan di database.

Storage

MinIO Local

S3 Production

Reason

Scalability.

Cost efficiency.

ADR-012

Use Audit Logging

Status

Accepted

Decision

Seluruh aktivitas penting wajib dicatat.

Examples

- Login
 - Status Change
 - User Update
 - Agency Update
-

ADR-013

Use Event Driven Internal Architecture

Status

Accepted

Examples

ReportCreated

↓

NotificationCreated

↓

AnalyticsUpdated

Reason

Future scalability.

ADR-014

Use OpenTelemetry + Sentry

Status

Accepted

Decision

Observability sejak awal.

Metrics

- Error Rate
 - Latency
 - Request Count
 - Database Performance
-

ADR-015

Use Docker First Deployment

Status

Accepted

Decision

Semua environment menggunakan Docker.

Reason

Environment consistency.

Deployment portability.